

## Research Assignment

A Database is a Systematic way of Storing and Managing data. It Comprises of Physical hardware in which the data is stored and also comprised of Software Component that organizes and Controls access to data. Databases justifies modern IT Infrastructure including Websites, platforms and applications such as Amazon and google

Real world examples of Where databases are used:

### 1. IN BANKS

→ Banks are using databases to process transactions, to update Client balances of accounts in real time and to prevent fraud, online Banking Sy

### 2. IN RETAILERS

→ Databases are used to maintain Massive Product Catalogues, to track available Inventory and to process Customer orders in Ecommerce platform

## Why databases are Essential for Modern world?

- They facilitate actionable analytics by organizing raw data, allows organizations to run complex queries, to uncover trends and to reliable data driven business decisions
- Every digital Service relies on databases to instantly process user requests, to track Inventory and personalize experiences
- Database ensure Information is accurate even when millions of users are accessing and updating the system simultaneously
- Databases allow Massive Scalability because they can expand accommodate billions of records without affecting the performance as user base grow



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## Datawarehouse Definition

- IS a large scale repository that combines and consolidate data from multiple sources to enable historical analysis, reporting and business intelligence
- It summarizes data from various sources into a central data store for querying and analysis
- Datawarehouse is a central repository of information that can be analysed to make informed business decisions

## How Datawarehouse is different from a database

### A Database

- Receives data instantly as users interact with an application/s

### A Datawarehouse

- Ingests data periodically in bulk collections often after cleaning it for analysis.

## Why Companies use a datawarehouse

- Companies use it to improve data driven business decisions and to improve organizational performance
- Companies use it to extract data from source systems, cleaning and transforming data and loading it to a data store

## 3. What is data LAKE

- Refers to a massive central location, a digital storage space that holds large amount of data (unprocessed data in its original format)

## 4. How Data Lake differs from datawarehouse

In Data lake, it is easy to dump everything immediately in the unsorted box



### In Data warehouse

- There is a Sorted Storage box with a highly organized filing Cabinet where every piece of data is Sorted by date, Category and Importance

### When would Business Prefer Data lake over Datawarehouse

- The business would prefer using data lake when it needs to store large volumes of raw unstructured or Semi Structured data at low cost
- Data lake approach will be preferred when the priority is to dump data into a Central repository immediately, when storing massive streams of data from Internet of things such as Sensor data

### 4. Data Mart Definition

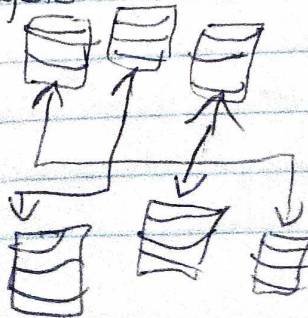
- is a Specialized Subset of a data warehouse focused on a specific functional area within an organization, providing a targeted Simplified view of data

#### How Data Mart relates to data Warehouse

Data Warehouse serves the whole organization, containing all historical and current operational data across multiple subjects therefore Data Mart acts as organizational Slice that draws information from the central data warehouse

Data Marts are used by departmental decision makers who need to generate quick reports, who needs to analyze trends and to track team specific Metrics

- Business users and analysts use data Marts





## 5. Structured DATA

- Refers to Information that is organized in a neat, Predictable format using fixed rows and Columns that Computers and humans can find immediately, Example is Excel Spreadsheet data

## 5.2 Semi-Structured DATA

- Refers to Information that is broken into identifiable tags or keys so that it can be searched. It does not use fixed rows and Columns
- Examples include Social Media posts, Sensor Data, Emails

## 5.3 Unstructured Data

- Refers to Information in its original format, It lacks predefined structure which makes it harder to search and analyze
- Examples are Video files, Audio files, emails, Photo media files

## 6. Data Modelling

- Refers to a Process of Creating a visual blueprint of how Information is Stored, Connected and organized in a system
- Data Modelling is essential because it Maps out how Information is organized, Stored and Connected, to Prevent repetition of data, to ensure high quality

### Types of Data Models

#### 1. Conceptual Data Model

- is a high level business focused view of data needed

#### 2. Physical Data Model

- Defines exactly how data will be stored physically in a specific database Management System



## 7. Data Mining

- Refers to a process of looking through massive amounts of data to find hidden trends, patterns or useful information
- Businesses use it to make operations effective and to improve profitability by studying customer behaviours and sales forecast

### Examples of Data Mining in Realtime

- Netflix and YouTube continuously mine user activity they analyze browsing history, previous purchases, search queries and the time of day you engage platform
- Amazon uses basket analysis to identify patterns in what items customers frequently buy

## 8. What is DBMS

- Refers to a software application that acts as the bridge between a database and its applications or users

### Popular Platforms of DBMS

#### 1. PostgreSQL

- Popularly known for its compliance, reliability and extensibility in SQL
- It handles complex data workloads

#### 2. MySQL

- Widely used in web development and online services
- Highly favoured for its speed, ease of use and compatibility

#### 3. Oracle

- Provides a multi-platform support
- Provides advanced analytics and security features

#### ③ MongoDB

- Stores data in flexible JSON like documents, making it SC
- It is ideal for unstructured, rapidly changing data



PostgreSQL

- Building ECommerce and Content Management System

Oracle

- In Global Financial and Banking Systems

MySQL

- Powering dynamic web applications such as Content Management System

MongoDB

ECommerce Product Catalog because it uses flexible document oriented schema

## 9. ETL

- Refers to a data Integration Process used to collect data from multiple sources, transforms it into a consistent format into a data warehouse or data repository

Extract

- Raw data is copied or exported from source locations to a staging area

Transform

- In staging area, Raw data undergoes data processing
- Data is transformed and consolidated for its intended Analytical use

Load

- transformed data is moved from staging area into a target data warehouse

## ETL

- Ensure data quality and Accuracy Fixing structural errors, removing duplicates, improved system performance by processing large amounts of data offline in staging areas before loading them into data warehouse



## 10. Common Storage Data Formats used in Analytics

### 10.1 Comma Separated Values

- text based tabular data format, in which each line represents a row of data and fields in a row separated by a comma
- Commonly used format for exporting spreadsheets or databases

### 10.2 JavaScript Object Notation

- lightweight text based data format for storing & sharing data which humans and computers can read easily

### 10.3 Avro Format

- It can handle data exchange efficiently between different systems and programming languages

### 10.4 Parquet File Format

- data files are efficient and their advanced compression technology minimizes storage needs



## SECTION B

### Artificial Intelligence & Machine Learning

#### 11. Artificial Intelligence

- Refers to the simulation of human intelligence in machines that are programmed to think and to learn like humans. These machines are designed to mimic cognitive functions such as problem solving, learning and pattern recognition. These tasks include visual perception, speech recognition, language translation and decision making.

##### Narrow AI

- Refers to machines or systems designed to do one specific job, these systems cannot think or learn outside its programmed task.

##### Examples

- Netflix recommendations

##### General AI

- Refers to machines or systems designed to learn, perform and understand different tasks like a human. It is a computer system with human level intelligence. It can plan, reason, learn any intellectual task a human can do.

##### Examples

- Driving a car
- Writing a program

#### 12. Machine Learning

- is simply a branch of Artificial Intelligence wherein computers learn using data to make decisions without being explicitly programmed. It is different from traditional programming by using data and outputs rather than using rules and data to find outputs.



## Types of Machine Learning

### ① Supervised learning

- Computer learns from labelled data, the correct answer is already provided

### ② Unsupervised learning

- Computer receives data without labelled answers and tries to find ~~new~~ patterns on its own. Eg grouping done by system according to customer behaviours

### ③ Reinforced learning

- Computer learns by trying different actions and receiving reward or penalties. Eg Robots learning how to move

## 13. Deep learning

- Refers to advanced computer technique that teaches software to process data and solve complex problems using systems inspired by human intelligence or brain

Relationship between Artificial Intelligence, Machine learning, Deep learning and Neural networks

### Artificial Intelligence

- └─ Makes Computers behave Intelligently

### Machine learning

- └─ Makes Computers learn from data

### Deep learning

- └─ Makes Computers learn using many layers

### Neural Networks

- └─ Refers to a system that learns patterns



## 14 Natural Language Processing

- IS A branch of Artificial Intelligence that allows Computers to Interpret, understand, Process and generate human language
- Helps Computers to understand the way humans Communicate

Real world applications:

- ① Google Search - Uses language understanding to determine what you seek
- ② Siri and Google Assistant - Speech recognition Converting Voice into text
- ③ AI Assistant & Chatbots - Helps the System understand what are you asking for and generate appropriate responses

## 15 Generative AI

- IS A type of Artificial Intelligence that can Create a new Content rather than analyzing existing information

Generative AI at high level

- ① Training
  - AI Model is trained using Very large amounts of data
- ② You provide a Prompt
  - You give an instruction to an AI
- ③ The model generates an Output
  - The model processes your Prompt and uses what it learned during Training to generate a response

## 17 Training Dataset

- IS A Collection of examples that is used to teach an AI how to recognize patterns and make predictions
- The quality of training data is important because an AI Model can only learn from the information given



## 18 Computer Vision

→ is a branch of artificial intelligence that enables computers to see, analyze and understand video and images

Step ① The ~~Camera~~ Captures an Image

Step ② The Computer represents the image as pixels

Step ③ The AI looks for Patterns

Step ④ The Model makes an Interpretation

## Two Real world use Case of Computer Vision

### ① Medical Image analysis

→ Computer vision can be used to analyze medical images such as X-rays

### ② Self Driving and Driver assistance systems

→ Cars use cameras and computer vision to identify the surroundings

## Section C: API, Tools & Developer Concepts

### 19. Application Programming Interface (API)

→ is a way of two different software systems of communicating with each other

① A Waiter in a restaurant may act as an API

You make a request → The waiter takes your request to the kitchen

↓  
The kitchen prepares the food

The waiter brings the food

② The Shopping Application communicates with a bank through API  
Shopping API → Payment API → Bank System → Payment Received

↓  
Shopping App



## 20. Api key

- IS a Unique String of Characters that an application uses to identify itself when communicating with an Application Programming Interface

### - Api key Purpose

- ① Identify the application making API Requests
- ② Control access to an API
- ③ Monitor API usage
- ④ Apply quotas and rate limits
- ⑤ Associate API usage with a particular Project
- ⑥ Restrict how an API can be used

### How an API key can be stored and protected

- API key must be treated like a password
  - It gives an application a permission to access an API
- ① Use environment process - store the key outside the source code
  - ② Use Secret Manager - Systems that encrypt and control access to sensitive
  - ③ Never put API keys directly in code
  - ④ Don't commit keys to Github
  - ⑤ Restrict access
  - ⑥ Rotate keys regularly

## 21. Rest API

- Refers to the way that different software applications communicate with each other over the Internet using standard HTTP requests

Rest stands for Representational State Transfer



A mobile banking application uses Rest API to Communicate with the bank Server

Mobile App  $\rightarrow$  Rest API  $\rightarrow$  Bank Server  $\rightarrow$  Rest API  $\rightarrow$  Mobile App

Rest is an architectural style for designing web services

Rest API uses  $\rightarrow$  URL's to Identify resources

$\rightarrow$  HTTP Methods to Specify what action could be performed

$\rightarrow$  JSON to exchange Data

$\rightarrow$  HTTP Status code to Indicate whether a request Succeeded

Main HTTP Methods used in Rest API

Get - Retrieves data

Post - Creates data

Put - Replace or update data

Delete - Removes data

## ② JSON (Javascript Object Notation)

- It is a lightweight format used to store and exchange data between applications

- It is language Independent. Can be used with Python, Java, C#, SQL

JSON is commonly used with API for data exchange because

- ① It is Easy to read - humans can understand the structure easily
- ② Easy for Computers to process - Programming languages can quickly parse JSON
- ③ Lightweight - Requires less data than complex formats
- ④ Language Independent - Different Programming languages can communicate using JSON

② Prompt Engineering Refers to the process of designing clear and effective instructions for an AI system so it produces useful response



Prompt Engineering is important because the quality of the prompt can affect the quality of the AI response.

A good prompt can

- Give the AI a clear understanding of what you want
- Provide necessary context and information
- Specify required length and level of detail

## (24) A Map Function

- Refers to a programming function used to apply the same operation to every item in a collection such as array.

In programming instead of writing a loop to process each item individually, Map allows you to transform all items in a simple way.

### Python Example

```
numbers = [1, 2, 3, 4, 5]
```

```
result = list(map(lambda x: x * 2, numbers))
```

```
print(result)
```

Output is

```
[2, 4, 6, 8, 10]
```

### Map in data processing

In data processing, `Map()` can be useful for transforming data.



## 25) Cloud Computing

- Refers to delivery of Computing Services over the Internet instead of running everything on physical Servers

### 3 Main Cloud Service Models

#### ① IaaS Infrastructure As A Service

- Provide Virtual Servers, Storage and networking
- The customer manages the operating system and applications

#### ② Platform As A Service

- Provide platform and tools for developers to build, test and deploy applications without managing underlying infrastructure

#### ③ Software As a Service

- Provides ready to use software over the Internet
- The Cloud provider manages the application and infrastructure

## 26) Version Control System

is a tool that tracks changes made to files over time

- It allows developers and data analysts to see what was changed

### Git

- is a popular distributed Version Control System used to manage and track changes in code, SQL scripts and notebooks

### Commit

- Refers to a saved snapshot of changes made to a project

### Branch

- Refers to a separate line of development within a repository



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